MODEL LAB · 2026-10-24

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

20,133
held-out plate appearances

2.97%
held-out HR rate

0.6009
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-10, scored on the 20,133 that came after (actual HR rate 2.97%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5832	1.51×	0.02872	0.13267	0.00563
KNN comps Prior is the pooled rate of the nearest comparables.	0.6009	1.63×	0.02868	0.13158	0.00464
SVR Support-vector regression on the same feature matrix.	0.5952	1.46×	0.02871	0.13232	0.00600
Random forest Forest regression; a learned similarity metric.	0.5980	1.54×	0.02869	0.13175	0.00464
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6009	1.63×	0.02868	0.13158	0.00464
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6009	1.63×	0.02868	0.13158	0.00464
Linear regression Ordinary least squares on the same features.	0.5974	1.57×	0.02869	0.13193	0.00514
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5975	1.56×	0.02871	0.13204	0.00552
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5976	1.63×	0.02869	0.13188	0.00543
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5960	1.56×	0.02870	0.13212	0.00552

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **257** knn **257** svr **257** rf **257** core **257** ensemble **257** linear **257** logistic **257** nn **257** xgb **257**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,884 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6009	1.09×	0.02383	0.11484	0.01705
KNN comps Prior is the pooled rate of the nearest comparables.	0.6332	2.18×	0.02370	0.11208	0.01018
SVR Support-vector regression on the same feature matrix.	0.6097	1.52×	0.02393	0.11661	0.01831
Random forest Forest regression; a learned similarity metric.	0.6353	1.96×	0.02372	0.11209	0.01174
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6332	2.18×	0.02370	0.11208	0.01018
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6332	2.18×	0.02370	0.11208	0.01018
Linear regression Ordinary least squares on the same features.	0.6390	1.96×	0.02375	0.11327	0.01360
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6346	1.96×	0.02380	0.11321	0.01386
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6312	2.18×	0.02372	0.11258	0.01436
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6196	1.31×	0.02388	0.11478	0.01757

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Junior Caminero 15.9%, Jackson Chourio 15.0%, Jonathan Aranda 14.7%, Jake Bauers 14.0%, William Contreras 13.8%, Yandy Díaz 12.9%
Flat league	6/6	Junior Caminero 15.3%, Jackson Chourio 15.2%, Jonathan Aranda 14.1%, William Contreras 13.7%, Jake Bauers 13.3%, Yandy Díaz 13.2%
KNN comps	6/6	Junior Caminero 15.8%, Jackson Chourio 15.6%, Jonathan Aranda 14.8%, Jake Bauers 14.5%, William Contreras 13.8%, Yandy Díaz 13.1%
SVR	6/6	Junior Caminero 17.1%, Jackson Chourio 15.9%, Jonathan Aranda 14.5%, Jake Bauers 14.3%, William Contreras 13.7%, Yandy Díaz 13.5%
Random forest	6/6	Junior Caminero 16.1%, Jackson Chourio 15.9%, Jonathan Aranda 14.6%, Jake Bauers 14.6%, William Contreras 13.7%, Yandy Díaz 13.2%
Core (KNN+SVR)	6/6	Junior Caminero 15.8%, Jackson Chourio 15.6%, Jonathan Aranda 14.8%, Jake Bauers 14.5%, William Contreras 13.8%, Yandy Díaz 13.1%
Forest blend	6/6	Junior Caminero 15.8%, Jackson Chourio 15.6%, Jonathan Aranda 14.8%, Jake Bauers 14.5%, William Contreras 13.8%, Yandy Díaz 13.1%
Linear regression	6/6	Junior Caminero 16.0%, Jackson Chourio 15.7%, Jonathan Aranda 14.8%, Jake Bauers 14.2%, William Contreras 13.7%, Yandy Díaz 13.1%
Logistic regression	6/6	Junior Caminero 16.4%, Jackson Chourio 15.7%, Jonathan Aranda 14.5%, Jake Bauers 14.3%, William Contreras 13.8%, Yandy Díaz 13.0%
Neural net (torch)	6/6	Junior Caminero 16.4%, Jackson Chourio 15.7%, Jonathan Aranda 14.6%, Jake Bauers 14.4%, William Contreras 13.8%, Yandy Díaz 13.0%
XGBoost	6/6	Junior Caminero 16.7%, Jackson Chourio 15.8%, Jonathan Aranda 14.7%, Jake Bauers 14.5%, William Contreras 13.6%, Yandy Díaz 13.4%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Junior Caminero**	TB	**15.9%**	15.3%	15.8%	17.1%	16.1%	15.8%	15.8%	16.0%	16.4%	
2	**Jackson Chourio**	MIL	**15.0%**	15.2%	15.6%	15.9%	15.9%	15.6%	15.6%	15.7%	15.7%	
3	**Jonathan Aranda**	TB	**14.7%**	14.1%	14.8%	14.5%	14.6%	14.8%	14.8%	14.8%	14.5%	
4	**Jake Bauers**	MIL	**14.0%**	13.3%	14.5%	14.3%	14.6%	14.5%	14.5%	14.2%	14.3%	
5	**William Contreras**	MIL	**13.8%**	13.7%	13.8%	13.7%	13.7%	13.8%	13.8%	13.7%	13.8%	
6	**Yandy Díaz**	TB	**12.9%**	13.2%	13.1%	13.5%	13.2%	13.1%	13.1%	13.1%	13.0%	
7	**Brice Turang**	MIL	**12.9%**	13.0%	12.6%	12.7%	12.4%	12.6%	12.6%	12.2%	12.1%	
8	**Victor Mesa Jr.**	TB	**12.2%**	12.0%	12.3%	12.8%	12.6%	12.3%	12.3%	12.5%	12.4%	
9	**Garrett Mitchell**	MIL	**11.8%**	11.4%	11.9%	10.7%	11.4%	11.9%	11.9%	11.4%	11.3%	
10	**Christian Yelich**	MIL	**11.5%**	11.7%	11.8%	11.2%	11.7%	11.8%	11.8%	11.9%	11.3%	
11	**Liam Hicks**	TB	**8.3%**	9.4%	8.4%	9.0%	8.6%	8.4%	8.4%	8.7%	8.5%	
12	**Jonny DeLuca**	TB	**7.6%**	8.1%	7.0%	7.2%	6.9%	7.0%	7.0%	7.2%	7.5%	
13	**Joey Ortiz**	MIL	**6.1%**	7.5%	6.5%	6.6%	6.7%	6.5%	6.5%	6.5%	6.5%	
14	**Sal Frelick**	MIL	**6.0%**	7.7%	6.2%	5.5%	5.6%	6.2%	6.2%	5.9%	6.0%	
15	**Cooper Pratt**	MIL	**5.8%**	7.5%	5.8%	5.3%	5.5%	5.8%	5.8%	5.0%	5.2%	
16	**Ben Williamson**	TB	**5.3%**	8.5%	6.0%	5.1%	5.0%	6.0%	6.0%	3.8%	5.5%	
17	**Taylor Walls**	TB	**5.2%**	6.7%	5.3%	4.0%	4.0%	5.3%	5.3%	5.0%	5.0%	
18	**Chandler Simpson**	TB	**3.0%**	6.4%	4.2%	1.7%	1.8%	4.2%	4.2%	1.7%	3.5%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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